

ELEMENT I

Our high-priority design requirements include The handle of the design, the dispensing mechanism of the design, and the button. Our low-priority parts of the design include the paint, excess wood, and fancy/extravagant designs on the outside of our box.

Testing:

Handle: We discussed different kinds of handles that would work well with our design. We decided to go with a premade handle rather than making one out of wood. If we made the handle out of wood it would've become an issue because it had the possibility of being off balance. This would've caused our project to not have the stability it needed to roll along different kinds of surfaces/terrain.

Dispensing Mechanism: We watched several videos of different kinds of dispensing toys or machines. This included (<https://www.youtube.com/watch?v=OkeSLuYtbD8>) the gumball machine dispense mechanism and (<https://www.drdishbasketball.com/why-dr-dish>) the basketball machine dispense mechanism. We studied both shooting mechanisms thoroughly. We noticed how powerful the basketball machine dispenser was and how efficient it has been for players for the past couple of years. This inspired us to make something that could really help the lacrosse players at ERHS. Both videos affected how we built our dispensing mechanism, but this video in particular <https://www.youtube.com/watch?v=aomavIH9Y64> was what truly inspired our dispensing mechanism. We followed the steps of how this dispenser was created and implemented it into our project. This video helped create the entire dispense mechanism overall.

Button: The button happened to be part of the dispensing mechanism but it was slightly different because part of it was going to be hanging outside of the dispenser. This meant that we had to come up with a design for the button that wasn't too rigid or ruff on the hands. We also had to size up the button to the scale of the dispenser so that it would effectively shoot out the lacrosse balls. Overall there is a common independent variable between all three high priorities; which is the amount of time we put in. The dependent variable for all of them is the effectiveness of the time we spent on each priority.



This is an image of us working on the inside of the dispenser. This contributed to the success of the dispensing mechanism.

<https://drive.google.com/file/d/1ZQ9peMbwdawEk9i44sb5LHIMmtkamWnd/view?usp=sharing>

This is a video part of the process of the construction as well. Overall we ended up coming up with good ideas for our high priorities because the dispenser worked just fine. Video of the function of the dispenser can be found shared in Google Drive from Korn Ngupkam . (There have been technical difficulties with uploading the link to the video) This is an image of the final overall design in which you can see the even handles and handy button design. The function of the dispense mechanism can be found in our skit video.



Based on our testing data we have concluded that the dispensing mechanism of the overall design could be improved. We noticed that the dispenser could get blocked up sometimes if the button wasn't pressed a certain way. In the future, we would probably change this by making the overall dispense mechanism inside of the dispenser have a smoother letdown.